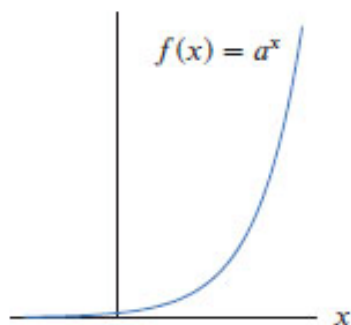


Sec 3.2 – The Exponential Function

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

What do we expect the graph of the exponential (**a is a constant and the variable is in the exponent**) $f(x) = a^x$ to look like?



$f(x)$ is always increasing, more slowly for $x < 0$ and more rapidly for $x > 0$. So the values of f' are small for $x < 0$ and large for $x > 0$.

In fact, the graph of the derivative of $f(x) = a^x$ is probably going to look like the graph of $f(x)$ itself!

The textbook has a nice derivation, but the rule for the derivative of an exponential function is:

Theorem

$$\frac{d}{dx} e^x = e^x$$

$$\frac{d}{dx} a^x = (\ln a) a^x$$

Note that **e** (the famous transcendental number, $e=2.7182\dots$) and **a** are constants, and the variable, **x** , is in the exponent.

$$\frac{d}{dx} 3e^x = 3e^x$$

$$\frac{d}{dx} 2^x + 5x^2 = (\ln 2)2^x + 10x$$

Partner up

Find the derivatives:

(a) $\frac{d}{dx} 5^x - 2x^5$

(b) $\frac{d}{dt} 5(3^t) + \sqrt{2}e^t - 7$

(c) $\frac{d}{dx} 3^{x+1}$

(d) $\frac{d}{dx} e^k + k^x$, where k is a constant

(e) $\frac{d}{dx} (\sqrt{2})^x - 3x^\pi$

Find the derivatives:

$$(a) \frac{d}{dx} 5^x - 2x^5 = (\ln 5)5^x - 10x^4$$

$$(b) \frac{d}{dt} 5(3^t) + \sqrt{2}e^t - 7 = 5(\ln 3)3^t + \sqrt{2}e^t$$

$$(c) \frac{d}{dx} 3^{1+x} = \frac{d}{dx} 3^1 3^x = 3(\ln 3)3^x$$

$$(d) \frac{d}{dx} e^k + k^x = 0 + (\ln k)k^x, \text{ where } k \text{ is a constant}$$

$$(e) \frac{d}{dx} \sqrt{2}^x - 3x^\pi = (\ln \sqrt{2})\sqrt{2}^x - 3\pi x^{\pi-1}$$

An animal population is given by $P(t) = 300(1.044)^t$, where t is the number of years since the study of the population began. Find $P'(5)$ and interpret your result. (3.2 #38)

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If $P(t) = 300(1.044)^t$, $P'(t) = 300(\ln 1.044)(1.044)^t$. Therefore, $P'(5) = 300(\ln 1.044)(1.044)^5 \approx 16.02$. This means that 5 years into the study, the population is growing by approximately 16 animals per year. (3.2 #38)